Operating System Concepts

- OS Concepts
- Linux commands
- Shell scripts
- Linux System call Programming

Learning OS

- step 1: End user
  - Linux commands
- step 2: Administrator
  - Install OS (Linux)
  - Configuration - Users, Networking, Storage, ...
  - Shell scripts
- step 3: Programmer
  - Linux System call programming
- step 4: Designer/Internals
  - UNIX & Linux internals

What is OS?

- Interface between end user and computer hardware.
- Interface between Programs and computer hardware.
- Control program that controls execution of all other programs.
- Resource manager/allocator that manage all hardware resources.
- Bootable CD/DVD = Core OS + Applications + Utilities
- Core OS = Kernel -- Performs all basic functions of OS.

OS Functions

- CPU scheduling
- Process Management
- Memory Management
- File & IO Management
- Hardware abstraction
- User interfacing
- Security & Protection
- Networking

Process Management

Program

- Set of instructions given to the computer --> Executable file.
- Program --> Sectioned binary --> "objdump" & "readelf".

- Exe header --> Magic number, Address of entry-point function, Information about all sections. (objdump -h program.out)
- Text --> Machine level code (objdump -S program.out)
- Data --> Global and Static variables (Initialized)
- BSS --> Global and Static variables (Uninitialized)
- RoData --> String constants
- Symbol Table --> Information about the symbols (Name, Size, section, Flags, Address) (objdump -t program.out)

* Program (Executable File) Format
    - Windows -- PE
    - Linux -- ELF
* Program are stored on disk (storage).

Process

* Progam under execution
* Process execute in RAM.
* Process control block contains information about the process (required for the execution of process).
    - Process id
    - Exit status
        - 0 - Indicate successful execution
        - Non-zero - Indicate failure
    - Scheduling information (State, Priority, Sched algorithm, Time, ...)
    - Memory information (Base & Limit, Segment table, or Page table)
    - File information (Open files, Current directory, ...)
    - IPC information (Signals, ...)
    - Execution context (Values of CPU registers)
    - Kernel stack
* PCB is also called as process descriptor (PD), uarea (UNIX), or task_struct (Linux).
* In Linux size of task_struct is approx 4KB

File Management

File

* File is collection of data/information on storage device.
    - File = Contents (Data) + Information (Metadata)
    - The data is stored in zero or more Data blocks (in FS), while metadata is stored in the FCB (in filesystem).
* FCB is called as "inode" on UNIX/Linux. It contains
    - type: UNIX/Linux has 7 types of files
        - -: regular, d: directory, l: symbolic link, p: pipe, s: socket, c: char device, b: block device
    - size: number of bytes
    - links: number of hard links
    - mode (permissions): (u) rwx, (g) rwx, (o) rwx
    - user & group
    - time-stamps: modification, creation, access.

    - info about data blocks
- terminal> ls -l
    - type, mode, links, user, group, size, timestamp, name.
- terminal> stat filepath

File System

- Files are stored on storage device. Arrangement of files in storage device is called as "File System".
- e.g. FAT, NTFS, EXT2/3/4, ReiserFS, XFS, HFS, etc.
- File System logically divide partition into 4 sections.
    - Boot block/Boot sector
        - Contains programs/info required for booting of OS
        - Typically contains bootstrap program and bootloader program
    - Super block/Volume control block
        - Contains information of whole partition.
        - Capacity, Label.
        - terminal> df -h
            - Total number of data blocks/inodes.
            - Number of used/free data blocks/inodes.
            - Information of free data blocks/inodes.
    - Inode List/Master file table
        - Inodes (FCB) for each file
    - Data blocks
        - Stores data of the file.
        - Each file have zero or more data blocks.
        - Size of data blocks can be configured while creating file system
- File system is created by the format utility while formatting the partition.
    - Windows: format.exe
    - Linux: mkfs
        - terminal> sudo mkfs -t ext3 /dev/sdb1
        - terminal> sudo mkfs -t vfat /dev/sdb1
        - -t fs_type e.g. ext3, ext4, vfat, ntfs, ...
        - partition e.g. /dev/sdb1
- Disk/partition naming conventions
    - Windows:
        - Disks are named as disk0, disk1, ...
        - partitions are named as drives i.e. C:, D:, E:, ...
    - Linux:
        - Disks are named as /dev/sda, /dev/sdb, /dev/sdc, etc.
        - Partitions per disk are named as
            - sda partitions: sda1, sda2, sda3, ...
            - sdb partitions: sdb1, ...

Linux File Structure

- Linux follows "/" (root) file system.
- "/" is a starting point of Linux file system.
- All your data is stored in this partition.
- / contains boot, bin, sbin, etc, root, home, dev, proc, mnt, media, opt
- In Linux everything is a file.
- Mainly there are two types of files in Linux
    - File
    - Directory (Folder)
- Linux Directories
    - boot - files related to booting
        - vmlinuz - kernel Image
        - grub - boot loader
        - config - kernel configuration
        - initrd/initramfs - initail root file system
    - bin - user commands in binary format
    - sbin - all admin/system commands in binary format
    - etc - configuration files
    - root - home directory of root user
    - home - it contains sub directories for each user with its name
        - devendra -> /home/devendra
        - sunbeam -> /home/sunbeam
        - osboxes -> /home/osboxes
    - dev - it contains all device related files
    - lib - shared program libraries required by kernel
    - mnt - it is temporary mount point
    - media - it is mount point for media eg cdrom
    - opt - stores optional files of large softwares
    - proc - virtual file system - it contains information about system or processes
    - sys - entries of each block devices, subdirectories for each physical bus type supported, every device class registered with the kernel, global device hierarchy of all devices
    - tmp - temporary files that may be lost on system shutdown
    - usr - read only directory that stores small programs and files accessible to all users

Path

- It is a unique location of any file in the file system.
- It is represented by character strings with few delimiters ("/", "\", ":")
- Types of path
    - There are two types of paths in linux
    - Absolute path
        - Path which starts with "/" is called as absolute path.
        - E.g. /home/devendra/MyData/Demos/demo01.sh

- Relative path
    - Path with respect to current directory is called as relative path
    - E.g. MyData/Assignments/assign02.pdf

Directory commands

- pwd -- print present working directory (current directory)
- cd -- change directory (syntax> cd dirpath)
- ls - list directory contents (syntax> ls dirpath)
- mkdir -- make directory (syntax> mkdir dirpath)
- rmdir -- remove empty directory (syntax> rmdir dirpath)
- pwd
    - pwd - print current working directory
- mkdir
    - mkdir dir - create directory
    - mkdir -p a/b/c - create nested directories
- rmdir
    - rmdir dir - remove empty directory
- cd
    - cd ~ - change working directory to home directory
    - cd - - change working directory to old working directory
    - cd .. - change working directory to parent directory
- ls
    - ls - list the contents of present working directory
    - ls path - list the contents of given path

File commands

- touch
    - if file does not exist, empty file is created
    - if file exist, timestamp of that file is changed
- rm
    - rm file - remove file
    - rm -r dir - remove directory with contents
    - rm -rf dir - force remove directory

- cat
    - cat file - display file content
    - cat > file - create new file
    - cat >> file - append to file

File/Directory copy and move

- mv
    - mv filepath destdirpath <-- move given file into given dest directory
    - mv dirpath destdirpath <-- move given dir into given dest directory
    - mv oldname newname <-- rename given file
- cp
    - cp filename newfilename <-- copy file with new name/path.
    - cp filepath destdirpath <-- copy file into given dest dir with same name.
    - cp -r dirpath destdirpath <-- copy file into given dest dir with same name.
- Special Characters
    - ~ - home directory
    - . - current directory
    - .. - parent directory
    - / - root directory separator
    - \ - escape character
    - ■ ■ matches multiple characters (wildcard)
    - ? - matches single character

Linux Admin

- In Linux, Admin is called as "super-user".
- Admin's login name is "root".
- Most of modern Linux, disable "root" login (for security).
- To execute commands with admin privileges use "sudo" (if approved by system admin).
    - cmd> sudo apt update
    - cmd> sudo apt install vim gcc python3 python3-pip
    - cmd> sudo snap install --classic code